

CSCE 649 PHYSICALLY-BASED MODELING · MODULE 09 · LECTURE NOTES

Deformable Solids

Continuum mechanics, and why mass-spring was never going to work.

MODULE OUTCOMES

- Define strain and stress and explain what each measures.
- Explain why linear elasticity fails under rotation.
- Implement corotational FEM.
- Handle element inversion.
- Choose between FEM, mass-spring, and shape matching.

1 Strain

ROTATION IS NOT DEFORMATION, AND THAT IS THE WHOLE DIFFICULTY

Take a rubber block and rotate it by 90°. Every vertex has moved a long way. **The material has not deformed in the slightest**, and no internal forces should arise.

So any strain measure must report exactly zero for a rigid motion — translation or rotation — and non-zero only for genuine stretching, compression, or shear.

Mass-spring models fail this test fundamentally. A spring measures only the distance between its endpoints, which carries no information about the configuration of the surrounding material. This is why a mass-spring cube can be turned inside out with every spring at its rest length: the springs are individually content and the material is destroyed.

Getting this right is precisely what continuum mechanics buys, and it is why this module exists.

$$\mathbf{F} = \partial \mathbf{x} / \partial \mathbf{X}$$

The **deformation gradient** $\mathbf{F}$ relates deformed positions $\mathbf{x}$ to material (rest) positions $\mathbf{X}$. For a tetrahedral element it is a 3×3 matrix computed from the current and rest vertex positions, and **everything else in this module derives from it**.

Strain measure	Definition	Under pure rotation
Cauchy (linear)	$\boldsymbol{\varepsilon} = \frac{1}{2}(\mathbf{F} + \mathbf{F}^T) - \mathbf{I}$	Non-zero. Wrong. Reports strain for a rotation, so the material resists being rotated.
Green (nonlinear)	$\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$	Exactly zero , because a rotation $\mathbf{R}$ satisfies $\mathbf{R}^T \mathbf{R} = \mathbf{I}$, so $\mathbf{F}^T \mathbf{F} = \mathbf{I}$ and $\mathbf{E} = 0$.

That identity is the entire trick. Rotations are orthogonal, so forming $F^T F$ annihilates them and leaves only the stretch. The cost is that Green strain is quadratic in the displacements, so the resulting system is nonlinear and requires an iterative solve rather than a single linear one.

THE LINEAR-ELASTICITY ARTEFACT EVERYONE HAS SEEN

Cauchy strain is linear in displacement, which makes the whole system linear: assemble a stiffness matrix once, solve one linear system per step. It is fast, simple, and well understood.

And it reports non-zero strain for a pure rotation, so the model generates internal forces resisting a motion that does not deform the material at all.

The visible result is ghost forces and volume growth. A character's arm rotating at the shoulder *inflates* — the classic linear-FEM artefact, immediately recognisable once you know to look for it, and present in a great deal of early deformable-character work.

Linear elasticity is perfectly appropriate for small deformations — a loaded beam deflecting a few millimetres, a vibrating plate — which is what it was developed for. It is useless for animation, where large rotations are the normal case.

2 Stress and constitutive models

Strain measures deformation; **stress** is the internal force per unit area that results. The relationship between them is the **constitutive model**, and it is where the material's identity lives.

Model	Character	Assessment
St. Venant–Kirchhoff	Linear stress computed from Green strain. The simplest model that is rotation-correct.	Collapses under compression. The energy is non-convex for compressed states, so a heavily squashed element is happy to collapse to zero volume. Not usable for large deformation.
Neo-Hookean	Nonlinear, derived from a hyperelastic energy with a term that diverges as volume approaches zero.	The usual choice. Resists compression strongly and correctly, behaves sensibly under large deformation, and has two parameters that map onto measurable quantities.
Mooney–Rivlin	More parameters, fitted to measured stress–strain curves.	Used when matching a specific real rubber matters. Rarely necessary in graphics.
Corotational	Linear elasticity with the rotation factored out (§3).	Fast, stable, and good enough for most animation. The practical default.

Parameter	Measures	Typical values
Young's modulus E	Resistance to stretching, in pascals.	Rubber $\approx 10^7$ Pa; soft tissue 10^4 – 10^6 ; steel $\approx 2 \times 10^{11}$.
Poisson's ratio ν	How much the material bulges sideways when compressed.	0.5 is perfectly incompressible; rubber ≈ 0.49 ; cork ≈ 0 ; most metals ≈ 0.3 .

These are real, published, measurable quantities, and that is the practical payoff over mass-spring models. You can look up the Young's modulus of silicone rubber. There is no published spring constant for silicone rubber, and there could not be, because it would depend on your mesh.

A caution: ν approaching 0.5 makes the material nearly incompressible, which is numerically difficult — the system becomes ill-conditioned and standard elements exhibit **locking**, becoming artificially stiff. Most practitioners use 0.45 and accept slight compressibility rather than fight it.

3 Corotational FEM

```
for (auto& e : elements) {
    mat3 F = compute_F(e, x);           // deformation gradient
    mat3 R = polar_decomposition_rotation(F); // F = R * S
    mat3 Fhat = R.transpose() * F;      // rotation removed
    mat3 P = linear_stress(Fhat);       // cheap model, now valid
    accumulate_forces(e, R * P);        // rotate forces back
}
```

POLAR DECOMPOSITION IS THE ENTIRE METHOD

Any deformation gradient factors uniquely as $\mathbf{F} = \mathbf{R} \cdot \mathbf{S}$, where $\mathbf{R}$ is a rotation and $\mathbf{S}$ is a symmetric positive-definite stretch. This is the **polar decomposition**, and it separates 'how the element has been turned' from 'how it has been deformed'.

Linear elasticity is accurate for small deformations when there is no rotation. Its only failure, as §1 established, is under rotation. So: factor out $\mathbf{R}$, apply the cheap linear model in the element's own rotated frame where it is valid, and rotate the resulting forces back into world space.

The cost is one polar decomposition per element per step — a few iterations of a simple fixed-point algorithm, or a closed-form expression in 3D. Cheap.

The result is linear elasticity's speed and simplicity with none of its rotation artefacts. The insight is almost embarrassingly simple, which is exactly why corotational FEM became the default for interactive deformables.

ELEMENT INVERSION MUST BE HANDLED, NOT PREVENTED

Under a sufficiently violent impact, a tetrahedron can turn inside out: its deformation gradient has **det $\mathbf{F} < 0$** .

Most constitutive models produce nonsense in that regime. The forces point in the wrong direction, driving the element further inverted, which produces larger wrong forces. The simulation fails within a few steps, and the failure propagates to neighbouring elements.

You cannot reliably prevent inversion — doing so would require bounding the forces the rest of the scene can apply, which is not under your control. So it has to be survivable.

Irving, Teran and Fedkiw's approach takes the singular value decomposition of $\mathbf{F}$, clamps the singular values away from zero, and **extends the constitutive model smoothly into the inverted regime** so that the forces push the element back out rather than further in. **Without this, a deformable simulation is one bad frame from failure; with it, inversion is recoverable and usually invisible.**

4 Alternatives

Method	Strengths	Weaknesses
FEM	Real material parameters. Behaviour converges as the mesh refines. Volume is represented. Anisotropy can be specified.	Substantially more complex. Requires a linear solver, and an implicit one for stiff materials.
Mass-spring	Trivial to implement; fast; adequate when art-directed.	Mesh-dependent behaviour; no volume preservation; parameters are not material properties (Module 03).
Shape matching	Unconditionally stable and very fast. Match the current point cloud to the rest shape with a best-fit rigid transform and pull toward it.	Not a material model at all. No concept of stress; stiffness depends on solver settings. This is to deformables what PBD is to constraints — theoretically weak and practically ubiquitous.
MPM	Handles topology change, plasticity, fracture, granular materials — snow, sand, mud. Module 12.	Expensive, and inherits grid artefacts from its Eulerian half.

THE CONCRETE CASE FOR FEM OVER MASS-SPRING

Parameters are real. Young's modulus for silicone is in a handbook. A spring constant for silicone is not, and cannot be.

Behaviour converges under refinement. A finer mesh gives more detail of the same material, rather than a different material (Module 03's three-resolution test).

Volume is represented. An element knows its own volume and knows when it has been inverted. A spring network has no such concept, which is why mass-spring solids collapse.

Anisotropy is specifiable rather than an accident of the triangulation — muscle fibre direction, wood grain, fabric warp.

The cost is a linear solve per step and considerably more code. For anything that must behave like a *particular* material, or that must survive mesh changes, it is worth it. For a gelatinous blob that only has to look fun, it is not.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Sifakis & Barbi — FEM Simulation of 3D Deformable Solids (SIGGRAPH course, free)	The best free introduction to this module. Notes and slides covering everything from the deformation gradient to implicit integration.
Müller et al. — Meshless Deformations Based on Shape Matching (free)	The shape-matching method of §4, from its authors.

Resource	What it covers for this module
Irving, Teran & Fedkiw — Invertible Finite Elements (2004, free)	The inversion handling of §3. Essential if you want a deformable simulation that survives contact.
Müller & Gross — Interactive Virtual Materials (free)	The corotational formulation of §3, clearly presented.
Bargteil & Shinar — Introduction to Physics-Based Animation	The deformables chapters tie §1–§4 together well.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Compute the deformation gradient for a tetrahedron and verify it is exactly the identity for a rigid translation and a rotation matrix for a rigid rotation.
- 2 Implement both Cauchy and Green strain. Apply a pure 45° rotation and report what each measure returns.
- 3 Implement linear FEM and rotate a bar by 90°. Measure the volume change and document the inflation artefact.
- 4 Implement polar decomposition and verify $F = R \cdot S$ with R orthogonal and S symmetric.
- 5 Implement corotational FEM and repeat the rotating-bar test. The volume change should now be negligible — report both numbers side by side.
- 6 Simulate the same object with Young's modulus spanning three orders of magnitude and confirm the behaviour matches expectation for the materials those values correspond to.
- 7 Vary Poisson's ratio from 0 to 0.49 and observe the bulging. Then try 0.499 and document the conditioning problem.
- 8 Force an element to invert by driving a collider through the object. Document the failure, then implement singular value clamping and show recovery.
- 9 Implement shape matching and compare against corotational FEM on the same mesh: stability, speed, and whether the parameters mean anything.
- 10 Run the three-resolution test from Module 03 on your FEM solid. It should pass where mass-spring failed.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why must a strain measure report zero for rigid motion, and why do mass-spring models fail this?
- 2 What is the deformation gradient, and what derives from it?

- 3 Why is Green strain zero under rotation? Give the one-line reason.
- 4 What artefact does linear elasticity produce, and when is it acceptable anyway?
- 5 Name four constitutive models and say when each is appropriate.
- 6 What are Young's modulus and Poisson's ratio, and why does it matter that they are measurable?
- 7 Explain corotational FEM in three steps.
- 8 What is polar decomposition and why does factoring out R make linear elasticity valid?
- 9 Why can element inversion not be prevented, and how is it handled?
- 10 Give four concrete advantages of FEM over mass-spring.